CoRoT-2b

R-1547

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_150625` · created 2026-08-01T19:40:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457198.8841 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.066
Transit depth (Rp/Rs)^2	0.00014 +/- 0.016 [%]
Orbital Inclination (inc)	86.43 +/- 2.09
Airmass coefficient 1 (a1)	0.62 +/- 0.41
Airmass coefficient 2 (a2)	0.38 +/- 0.95
Scatter in the residuals of the lightcurve fit is	66.97 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	5.5
Transit Duration (day)	0.077 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.066 vs 0.1667 (deviation 2.21σ)
Duration fit vs published	0.077 d vs 0.095 d (deviation 1.22σ)
Mid-time O–C	-6.2 min
Depth SNR	0.0
Residual scatter	66.97 %

Light curve

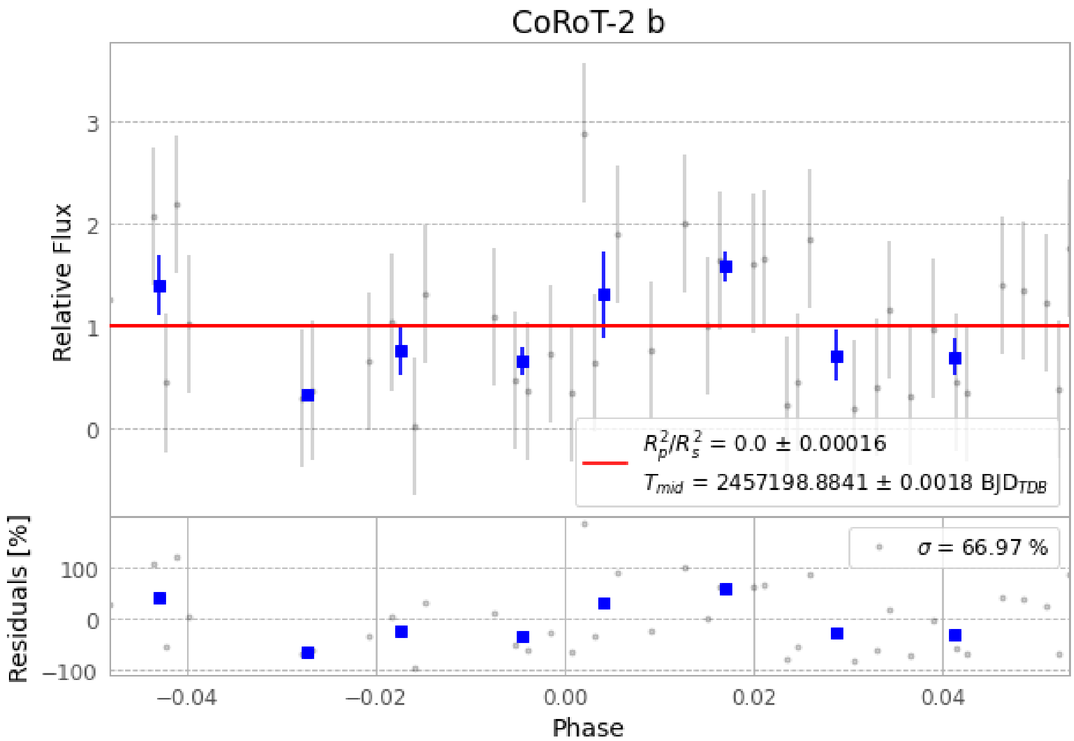

Figure 1 — FinalLightCurve_CoRoT-2 b_2015-06-25.png (SHA-256 f11d0168bcec1df3…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 414], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 aa9aba1b50ea264a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

86 FITS frames (56 MB), `CoRoT-2150625070312.FITS` → `CoRoT-2150625111812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-150625.log (32cdd3d1a3c5…), FinalLightCurve_CoRoT-2 b_2015-06-25.png (f11d0168bcec…), FinalLightCurve_CoRoT-2 b_2015-06-25.csv (3d23f4a33f39…)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)